

MA 587 Homework 01

Due: Friday 11:59pm, Feb 13, 2026

It should be submitted through Gradescope. **PLEASE ASSIGN PAGES to each problem** (making sure you're doing this correctly). **PLEASE make sure the file is in CORRECTLY ORIENTED.** Show all work and you **MUST** write up your own solutions.

1. (Mollifier) Let us consider a function defined by

$$\eta(x) = \begin{cases} e^{-\frac{1}{1-x^2}} & \text{if } |x| < 1 \\ 0 & \text{if } |x| \geq 1 \end{cases},$$

which is known as a mollifier. (Roughly speaking, a mollifier refers to a function that mimics the Dirac delta function. See more in this link). This problem aims at showing that η is a smooth function (infinitely many differentiable).

- (a) Show that η is continuous at $x = 1$ and $x = -1$.

Hint: Recall that the definition of “continuity”. Probably, a calculus-level definition is the one that many would be familiar with, which states that if f is continuous at $x = c$, then $\lim_{x \rightarrow c} f(x) = f(c)$. This definition works fine for this problem.

- (b) Show that η is infinitely many differentiable at $x = 1$.

For any $\epsilon > 0$, let $\eta_\epsilon(x) := \frac{1}{\epsilon} \eta(\frac{x}{\epsilon})$. If b and ϵ are chosen appropriately, it can be checked that $\eta_\epsilon(x - b)$ belongs to V . See the desmos visualization (demonstration-link).

2. (Fundamental Lemma of the Calculus of Variations - FLCV) Let

$$V = \left\{ v : \begin{array}{l} v \text{ is continuous on } [0, 1], \\ v' \text{ is piece-wise continuous and bounded on } [0, 1], \\ v(0) = v(1) = 0. \end{array} \right\}$$

Show that if w is continuous on $(0, 1)$ and $\int_0^1 w(x)v(x)dx = 0$ for all $v \in V$, then $w(x) = 0$ for all $x \in (0, 1)$.

Hint: You will need a rigorous definition of “continuous” for this problem. A real-analysis level definition states that f is continuous at $x = c$ if for any $\epsilon > 0$, there exists $\delta > 0$ such that for any $y \in (c - \delta, c + \delta)$, $|f(y) - f(c)| < \epsilon$.

For this problem, if $w(c) > 0$ for some $c \in (0, 1)$, by taking $\epsilon = \frac{w(c)}{2}$, the continuity definition gives an interval $(c - \delta, c + \delta)$ on which f is greater than $\frac{w(c)}{2}$. Note that δ depends on your choice of ϵ .

3. Consider the boundary value problem

$$\frac{d^4 u}{dx^4} = f \quad \text{in } (0, 1), \quad u(0) = u'(0) = u(1) = u'(1) = 0. \quad (1)$$

- (a) Show that the variational form of (1) is $\langle u'', v'' \rangle = \langle f, v \rangle$ for all $v \in W$, where

$$W = \left\{ v : \begin{array}{l} v \text{ and } v' \text{ are continuous on } [0, 1], \\ v'' \text{ is piecewise continuous,} \\ v(0) = v'(0) = v(1) = v'(1) = 0. \end{array} \right\}$$

(b) For an interval $I = [a, b]$, consider

$$P_3(I) = \{v : v \text{ is a polynomial of degree } \leq 3 \text{ on } I. \}$$

That is, any $v \in P_3(I)$ can be expressed as $v(x) = a_3x^3 + a_2x^2 + a_1x + a_0$ for $x \in I$ where $a_0, a_1, a_2, a_3 \in \mathbb{R}$.

Show that $v \in P_3(I)$ is uniquely determined by the values $v(a), v'(a), v(b), v'(b)$. That is, find four basis functions $\phi_{\text{left}}, \phi_{\text{right}}, \psi_{\text{left}}, \psi_{\text{right}} \in P_3(I)$ such that

$$\begin{cases} \phi_{\text{left}}(a) = 1 \\ \phi_{\text{left}}(b) = 0 \\ \phi'_{\text{left}}(a) = 0 \\ \phi'_{\text{left}}(b) = 0 \end{cases}, \quad \begin{cases} \phi_{\text{right}}(a) = 0 \\ \phi_{\text{right}}(b) = 1 \\ \phi'_{\text{right}}(a) = 0 \\ \phi'_{\text{right}}(b) = 0 \end{cases}, \quad \begin{cases} \psi_{\text{left}}(a) = 0 \\ \psi_{\text{left}}(b) = 0 \\ \psi'_{\text{left}}(a) = 1 \\ \psi'_{\text{left}}(b) = 0 \end{cases}, \quad \begin{cases} \psi_{\text{right}}(a) = 0 \\ \psi_{\text{right}}(b) = 0 \\ \psi'_{\text{right}}(a) = 0 \\ \psi'_{\text{right}}(b) = 1 \end{cases}.$$

Then, show that any $v \in P_3(I)$ can be expressed as

$$v(x) = c_1\phi_{\text{left}}(x) + c_2\phi_{\text{right}}(x) + c_3\psi_{\text{left}}(x) + c_4\psi_{\text{right}}(x),$$

where $c_1 = v(a), c_2 = v(b), c_3 = v'(a), c_4 = v'(b)$.

Hint: For example, the basis function corresponding to ϕ_{left} can be found as follows. Since ϕ'_{left} is a quadratic polynomial satisfying $\phi'_{\text{left}}(a) = \phi'_{\text{left}}(b) = 0$, we have

$$\phi'_{\text{left}}(x) = A(x-a)(x-b) \implies \phi_{\text{left}}(x) = A\left(\frac{1}{3}x^3 - \frac{a+b}{2}x^2 + abx\right) + C,$$

where A and C are determined by the conditions $\phi_{\text{left}}(a) = 1$ and $\phi_{\text{left}}(b) = 0$. It then can be checked that $A = \frac{1}{\frac{1}{3}(a^3-b^3) - \frac{a+b}{2}(a^2-b^2) + ab(a-b)}$ and $C = -A(\frac{1}{3}b^3 - \frac{a+b}{2}b^2 + ab^2)$, which gives

$$\phi_{\text{left}}(x) = \frac{(b-x)^2(2x-3a+b)}{(b-a)^3}.$$

Similarly, $\phi_{\text{right}}, \psi_{\text{left}}, \psi_{\text{right}}$ can be found.

(c) Consider a set of grid points $0 = x_0 < x_1 < \dots < x_M < x_{M+1} = 1$ and let $I_j = [x_{j-1}, x_j]$. By using basis functions from (b), construct a finite-dimensional subspace W_h of W consisting of piecewise cubic functions. Specify suitable parameters to describe the functions in W_h and determine the corresponding basis functions.

Hint: Let $\phi_{\text{left}}^{I_j}$ be the basis function ϕ_{left} defined on $P_3(I_j)$ and other functions are similarly defined. Consider the cubic-hat basis functions defined by

$$\varphi_j(x) = \begin{cases} \phi_{\text{right}}^{I_j}(x) & \text{if } x \in [x_{j-1}, x_j] \\ \phi_{\text{left}}^{I_{j+1}}(x) & \text{if } x \in [x_j, x_{j+1}] \\ 0 & \text{otherwise} \end{cases}, \quad \psi_j(x) = \begin{cases} \psi_{\text{right}}^{I_j}(x) & \text{if } x \in [x_{j-1}, x_j] \\ \psi_{\text{left}}^{I_{j+1}}(x) & \text{if } x \in [x_j, x_{j+1}] \\ 0 & \text{otherwise} \end{cases}.$$

(d) Formulate a finite element method for (1) based on the space W_h . Find the corresponding linear system of equations in the case of a uniform partition.

4. (Implementation of FEM for (1)) This exercise will guide you to implement the FEM for (1) in your favorite programming language (e.g. MATLAB or Python). The mesh is assumed to be equidistant, i.e., $h_j = h$ for all j . **You can use Gemini (or other ChatGPT variants) to complete this part. If so, please clearly state that you used Gemini or its variants) and provide a detailed prompt that generates the code. However, you must verify the correctness of the code before submission.**

- (a) Let $u(x) = \sin^2(\pi x)$ be the solution to (1). Find f .

The FEM implementation follows the four steps.

- (b) (Basis construction) Write a code that constructs the basis functions from P3(c).
(c) (Stiff matrix A) Write a code that constructs the stiff matrix A from P3(d).
(d) (Load vector b) Write a code that approximates the load vector b from P3(d) using the Simpson's quadrature rule.
(e) (FEM solution) Write a code that solves $A\xi = b$ and returns the FEM solution. That is, $u_h(x) = \sum_{j=1}^M \xi_j^* \phi_j(x)$ where ξ^* is the solution to $A\xi = b$ and ϕ_j 's are the basis functions from P4(b).
(f) (Error plot) Let X_{test} be the set of 10^5 equidistant points in $[0, 1]$. To quantify how accurate the FEM solution is, some quantity that indicates the accuracy of the solution needs to be introduced. The two popular choices are the relative ℓ_2 error and the maximum absolute error. These are defined as follows:

$$e_{\text{rel}}(h) = \sqrt{\frac{\sum_{x \in X_{\text{test}}} (u(x) - u_h(x))^2}{\sum_{x \in X_{\text{test}}} u^2(x)}}, \quad e_{\text{max}}(h) = \max_{x \in X_{\text{test}}} |u(x) - u_h(x)|.$$

By varying M from 5, 10, $\dots$, 100, provide the two error plots – One is the semilogy plot that shows the relative ℓ_2 errors with respect to the number M of basis functions. The other is the loglog plot that shows the maximum absolute errors with respect to the mesh size h . Note that $h = \frac{1}{M+1}$.